

## 1.1 What Is Thy Purpose?

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Imagine this: you've bought a shiny new PC, and you are quite happy with it—your favourite programs run just fine, and life in general looks good. Soon, of course, your PC will become too slow to run the latest versions of your applications, and you will need to upgrade. You've changed your motherboard and processor and now fire up the program, only to realize that it doesn't work! Believe it or not, this actually used to happen.

Programming for a computer usually meant knowing all the details about the hardware, and should that hardware change, programs would become useless and would have to be rebuilt to suit the new hardware. This was not so bad for early computer systems that were designed for just one or two purposes, but with the growing complexity of programs, it became necessary that programs be scalable and runnable on all computers, without having to bother about changing hardware.

The Operating System (OS) acts as a middleman between applications and hardware, with the help of which programmers don't need to know the nitty-gritty of the hardware they write codes for. Their programs just need to talk to the OS, which in turn talks to the hardware. To achieve this, the OS provides programmers with an *Application Program Interface* (API), which lets them write programs that can communicate with the OS.

Your computer has many *resources* that help applications achieve their purpose. The most important, of course, is the CPU. There is also memory, hard drives, network connections, and several other pieces of hardware. The most important function of an OS is to manage all these resources and make sure that each application gets its fair share of all the resources it needs—they should get enough of the CPU's time to get their instructions processed, enough memory to store their data in, and should be able to talk to any piece of hardware it wants to.